

Confirmation of clonal reproduction of *Fagus crenata* Blume from Sado Island, Niigata Prefecture

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Abstract

While clonal growth is important in the East Asian *Fagus* subgenus *Engleriana* and the North American *Fagus grandifolia* (subgenus *Fagus*), for other subgenus *Fagus* species the vast majority of regeneration involves sexual reproduction with clonal growth only rarely observed. Here we aim to confirm using nuclear microsatellite markers whether clonal growth occurs in the Japanese endemic species *Fagus crenata* (subgenus *Fagus*) by investigating the origin of multi-stemmed clumps found within a high-elevation dwarf beech forest on Sado Island, Niigata Prefecture. We found that all stems collected from three separate clumps belonged to the same clump-specific multi-locus genotypes. The maximum size of clones was 3–4 m in diameter, comparable in size to those seen in the predominantly asexually regenerating Japanese species *Fagus japonica* (subgenus *Engleriana*). The species capacity for clonal growth is likely to underlie its ability to persist at high-elevation exposed sites at the limits of its ecological range.

KEYWORDS

clonal growth, dwarf beech forest, *Fagus crenata*, nuclear microsatellites, Sado Island

1 | INTRODUCTION

A large percentage of plants have the capacity for clonal growth (Eckert, 1999), which is defined as the vegetative proliferation of ramets which may achieve independence and survive fragmentation from the original plant (Mogie & Hutchings, 1990). The importance of clonal growth in plants varies widely from cases where species rely solely on it for reiteration while in other species clonal growth only manifests itself after damage to the main stem or in extreme ecological conditions. In the important northern hemisphere temperate tree genus *Fagus* the prevalence of clonal growth varies being common, and sometimes the predominant mode of reproduction, in the three East Asian *Fagus* subgenus *Engleriana* species (Cao & Peters, 1998; Ohkawa, Kitamura, Takasu, & Kawano, 2006; Ohkubo, 1992) and the North American

subgenus *Fagus* species *Fagus grandifolia* (Jones & Raynal, 1986; Kitamura et al., 2000). However, in all other species of subgenus *Fagus* the majority of regeneration involves sexual reproduction, with clonal growth only having rarely been observed. Here we aim to clarify whether clonal growth has a role in the regeneration of the Japanese endemic species *Fagus crenata* (subgenus *Fagus*) by investigating the origin of multi-stemmed clumps of the species found on Sado Island, Niigata Prefecture (Figure 1) using nuclear microsatellite markers.

Fagus crenata is a dominant tree of cool temperate forests distributed from southern Kyushu to southern Hokkaido. On the Japan Sea side of Honshu, where snow cover is deep, the species can occur across a wide elevational range from the foothills over 200 m above sea level (masl) to elevations over 1700 masl (Tanaka & Taoda, 1996). While the regeneration of *F. crenata*

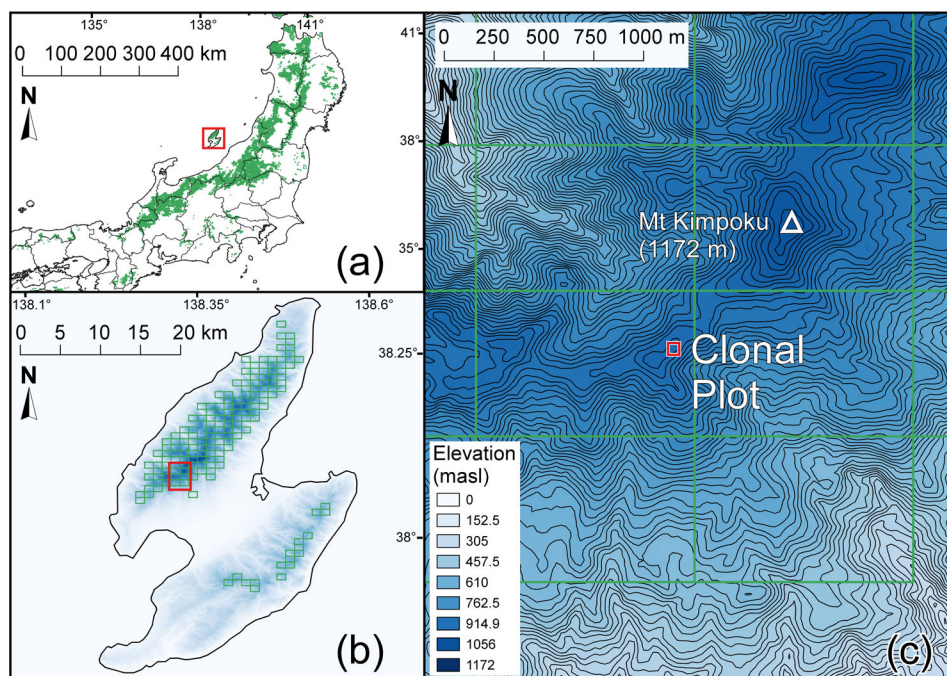


FIGURE 1 The location of the clonal plot on Sado Island, Niigata Prefecture (c) with the clonal plot's location in relation to Sado Island and mainland Honshu shown in insets (a) and (b). The plot occurs on a wide ridge at ~950 m above sea level on the main mountain range of the island. The distribution of *Fagus crenata*-dominated forest (sourced from Matsui, Yagahashi, Nakaya, Tanaka, & Taoda, 2004) is demarcated by green polygons for all panels. The contour interval was 15 m for panel (c)

involves almost exclusively sexual reproduction on both the snowy Japan Sea side and the low-snow cover Pacific Ocean side (Shimano, 2006), at the most extreme limit for growth of the species on wind-exposed ridges and summits with deep snow cover clonal growth has been reported (Homma, 1997; Tanimoto, 1993). In these dwarf beech forests the occurrence of multi-stemmed clumps is thought to be due to sprouting from near the root collar and layering (Tanimoto, 1993) associated with deformation of trunks by wind and heavy snow (Homma, 1997).

This study aims to confirm using nuclear microsatellites whether multi-stemmed clumps of *F. crenata* are in fact derived via clonal growth. To do this we investigate three clumps from a ridge-top dwarf beech forest on the Osado Mountain Range of Sado Island, where multi-stemmed individuals are common. We use highly polymorphic nuclear microsatellites, which are a powerful tool to investigate clonality, because, unlike studies based on visual inspection of clones, they can discriminate as to whether individual stems, whether physically connected or not, are derived via clonal growth or sexual reproduction with near complete accuracy. Nuclear microsatellite markers can also provide insights into the spatial distribution, size and admixture of clonal individuals and ramets.

2 | MATERIALS AND METHODS

2.1 | The site

The site where the multi-stemmed clumps of *F. crenata* are found is located between 948 and 957 masl on a

southwest–northeast directed wide flat ridge on the Osado Mountain Range of Sado Island, Niigata Prefecture (Figure 1). At this elevation the maximum snow depth is about 3 m and the snow season lasts for five and a half months, from mid-December to the end of May (Nakano & Sakio, 2018). The vegetation consists of an uneven canopy of spaced *F. crenata* and *Quercus crispula* individuals and an understory of *Clethra barbinervis*, *Ilex crenata* var. *paludosa*, *Rhododendron brachycarpum* and *Sasa* sp. Vegetation of the adjacent northwest and southeast slopes consists of a closed canopy of single-stemmed *F. crenata* forest. The site has not experienced any obvious artificial disturbance associated with road construction or logging.

2.2 | Sampling

Three multi-stemmed clumps of *F. crenata* in the stand were targeted for investigation of potential clonal growth (Figures 2 and 3). For each target clump a single GPS position was recorded. Given the closeness of some stems and the large number of stems (over 20 in two of three clumps, Table 1), leaf samples were collected from a subsample of stems with the objective to sample across the spatial extent of each clump and to ensure the complete circumference of the clump was represented. When two stems were immediately adjacent the stem with the largest diameter at breast height (DBH) was sampled. In this way 33% to 66% of all stems were sampled from each clump consisting of 6 to 9 samples (total 23 stems sampled; Table 1). For each sampled stem the diameter

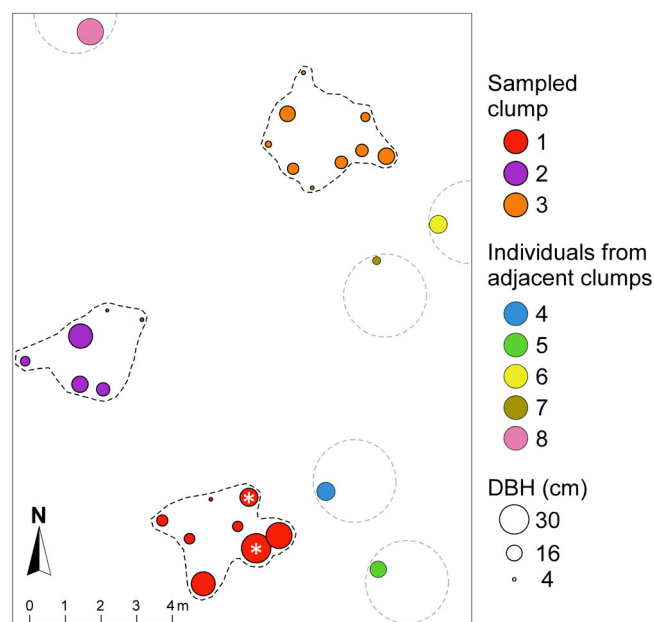


FIGURE 2 A map of the position in the plot of each multi-locus genotype identified using 12 nuclear microsatellite markers for the three target multi-stemmed clumps of *Fagus crenata* and individual stems sampled from adjacent clumps. The stem diameter at breast height (cm) is indicated by the size of the circles. The black stippled line indicates the edge of each target clump while the gray stippled circles indicate the approximate location of the adjacent clumps. The asterisks indicate the two stems of clump 1 with somatic mutations

(cm) was measured at 1.3 m height or, as some stems were leaning (sometimes at acute angles nearly along the ground) the diameter was measured at 1.3 m along the stem from where the stem left the ground. There was no threshold on the DBH of stems sampled but stems under 3 cm were rare and were not sampled in this study. In addition, the distance (m) of each sampled stem to the estimated center of each clump was measured with a measuring tape at the point where DBH was measured and the position of stems in relation to the clumps' estimated center was measured using a compass. Lastly, the nearest stems of adjacent multi-stemmed *F. crenata* clumps were sampled to determine whether stems outside the three targeted clumps belonged to different genotypes. This resulted in the sampling of two adjacent clumps nearest to clump 1, none to clump 2 (which was not close to any other individuals) and three to clump 3.

2.3 | Molecular methods

DNA extractions were undertaken using the Qiagen DNeasy Plant Mini Kit (Qiagen) following the manufacturers' instructions. Twelve nuclear microsatellite loci (FS103, FS446, mfc2, mfc12, sfc161, sfc36, sfc1063, sfc1105, sfc1143, sfc7, sfc378 and sfc18) used by (Kitamura et al., 2015) were used to genotype each individual. These markers are highly useful for identifying

FIGURE 3 Photos of the three target multi-stemmed clumps of *Fagus crenata* (a–c) and the single-stemmed forest dominated by *F. crenata* on the nearby northwest facing slope below the ridge (d)

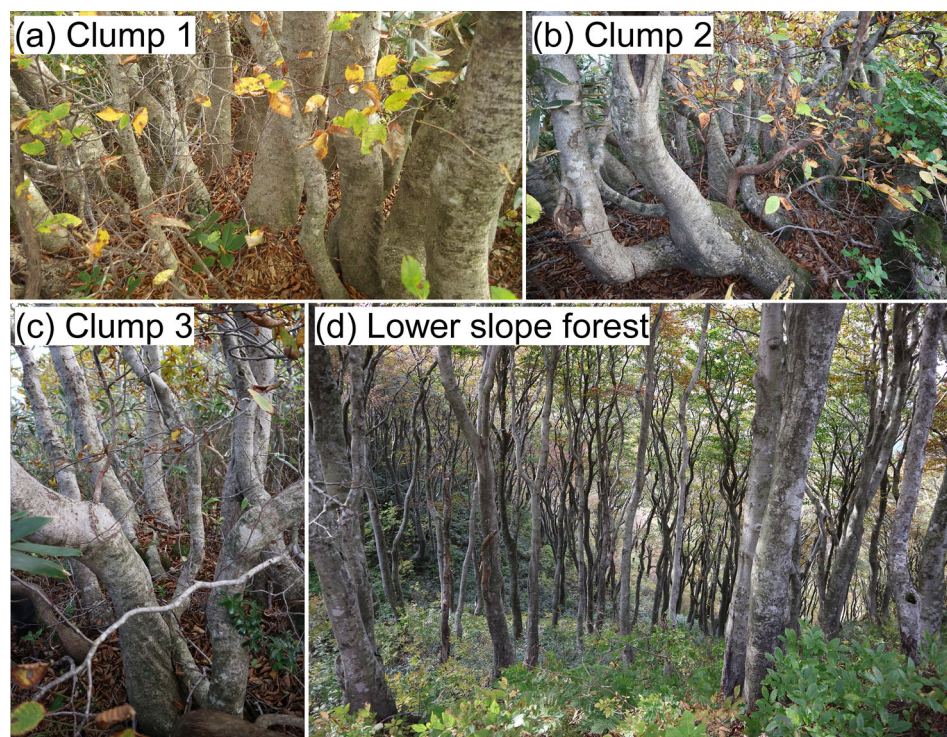


TABLE 1 Information on the target multi-stemmed clumps of *Fagus crenata* used to investigate clonal growth

Target multi-stemmed clump	No. of stems in clump	No. of sampled stems	Average DBH (cm) (standard deviation)	Min. DBH (cm)	Max. DBH (cm)
1	24	8	17.03 (9.05)	3.98	29.66
2	9	6	12.06 (7.95)	3.44	24.32
3	22	9	10.62 (4.73)	4.14	16.9

clones with a very low probability of identity (i.e., the probability that two unrelated individuals have the same genotype) of <0.0001 when calculated using 47 individuals collected from a sexually reproducing *F. crenata* stand from the Shirakami Range, Aomori Prefecture (unpublished data). The PCR thermocycle conditions followed (Kitamura et al., 2015). The length of amplified products was analyzed using an ABI 3130xl Genetic Analyzer with a GeneScan 600 LIZ size standard and fragments were scored using GeneMapper (Applied Biosystems, Inc.).

2.4 | Data analyses

Firstly, we investigated whether any sampled stems in the plot belonged to the same genotype (hereafter referred to as a multi-locus genotype [MLG]) using all samples (i.e., the three targeted clumps and adjacent clump samples) in GenClone 2.0 (Arnaud-Haond & Belkhir, 2007) using the “matrix of allele distance” function. This function enables the examination of the frequency distribution of the pairwise allele distance (Chakraborty & Jin, 1993) between all MLG pairs. This analysis indicated that two stems from clump 1 differed from the MLG of all other clump 1 stems by a single allele at one locus each (mfc2 and sfc1143; for the genotypes of all sampled individuals see Table S1) and, thus, were considered to be derived by somatic mutation (Ally, Ritland, & Otto, 2008). In order to estimate the likelihood that these two stems may in fact be derived from sexual reproduction we followed the procedure of Arnaud-Haond, Duarte, Alberto, and Serrao (2007), where by P_{sex} , that is, the probability that identical MLGs are derived from separate sexual reproductive events, was calculated using only the ten identical loci. If P_{sex} values were below 0.01, then the two stems were considered as being derived via asexual reproduction from the same MLG.

In addition, for all samples the clonal subrange was calculated in GenClone 2.0. The clonal subrange provides an estimate of the probability of clonal identity with increasing spatial distance between sampled stems. The distance where probability of identity approaches zero

(Harada, Kawano, & Iwasa, 1997) represents the spatial scale beyond which clonality does not impact the genetic structure of the stand (Arnaud-Haond et al., 2007). In addition, in the same program, an aggregation index was calculated by comparing the probability of clonal identity between nearest neighbors relative to that between pairs of samples chosen at random from the stand (Arnaud-Haond et al., 2007) using 1,000 permutations. The index varies between 0, indicating high intermingling of MLGs, and 1, indicating high aggregation. For the calculation of this index, separate analyses were undertaken using all samples and only the three target clumps. For both clonal subrange and aggregation index analyses, the two stems derived via somatic mutation from clump 1 (see Results below) were considered as the same MLG as other clump 1 stems.

3 | RESULTS

The average DBH of sampled stems for the three target clumps was between 10.6 and 17 cm with maximum diameters 4.1 to 7.5 times the size of the smallest stem (Table 1). We found that all stems collected from each of the three target clumps belonged to the same clump-specific MLGs consisting of 6 to 9 ramets (Figure 2). The two stems from clump 1 found to differ by one allele at one locus each from the MLG of all other clump 1 stems (Figure 4) were confirmed to be derived by somatic mutation. P_{sex} , calculated when using only the ten identical loci, was found to be <0.0001 (results not shown), well below the threshold value (0.01). All single stems collected from adjacent clumps were found to be distinct MLGs (Figure 2).

The maximum distance between stems of the same MLG was 4.09 m. The probability of genotypic identity was found to be 1 between 1–2 m distance, declining to 0.83 at 3 m, less than half (0.43) at 4 m, and reaching near zero beyond 5 m (Table 2). The aggregation index calculated when using all sampled stems was 0.82, which was highly statistically significant versus random expectations ($p < 0.0001$). When examining only the three target clumps the aggregation index was 1 ($p < 0.0001$).

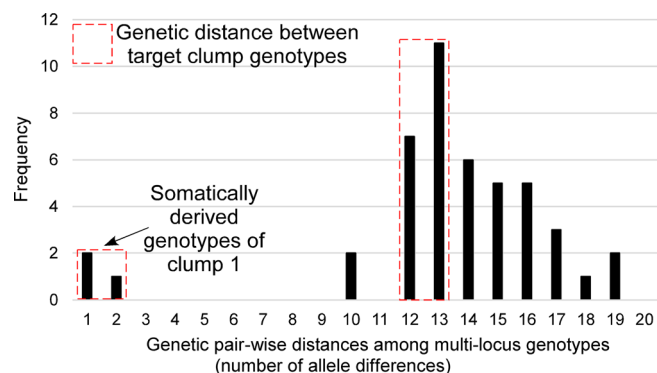


FIGURE 4 Frequency distribution of the pairwise allele distance between all multi-locus genotypes observed in both the three target clumps and single stems sampled from adjacent clumps including the two somatically derived genotypes observed in clump 1

4 | DISCUSSION

This study has confirmed using genetic markers that multi-stemmed clumps of *F. crenata* on Sado Island are derived via clonal growth. This is the first confirmation of clonal growth in the species and demonstrates that clonal growth can occur in all species of *Fagus* subgenus *Fagus* in Europe (Peters, 1997; Erlas, 2001), North America (Jones & Raynal, 1986; Kitamura et al., 2000) and Japan.

While not all stems in each target clump were included in the analysis, it is most likely that all stems belonged to the same clone because the high aggregation index indicates that the likelihood of admixture of ramets of different MLGs is very low. This is supported by the finding that no stems from the adjacent clumps were the same MLG as any target clump. Although our measuring method did not allow a completely accurate assessment of clone size, the maximum diameter of clones observed at the site was 3–4 m (Figure 2). This shows that *F. crenata* clonal clumps can reach a comparable size to those seen in the predominantly asexually reproducing Japanese subgenus *Engleriana* species *Fagus japonica* (Ohkubo, 1992). Given the size of the clones and the fact that in none of the clones a central main stem persisted, they are possibly hundreds of years old. However, it is difficult to evaluate the age of the individual clonal clumps because clone size is not a reliable indicator of when a clone was first established (Ally et al., 2008).

The mode of clonal growth, that is, whether from adventitious buds near the root collar (i.e., where the stem and the root system meet (Stone & Stone, 1954), root sprouting, a lignotuber or layering, is uncertain because we did not undertake excavations of the soil

TABLE 2 The clonal-subrange results using all samples collected from the plot including the distance classes examined, the probability of clonal identity (defined as the fraction of all the pairs of ramets sharing the same multi-locus genotype (MLG) and that are separated by a distance belonging to that interval (Harada et al., 1997)) and the number of pairs of stems within each distance class

Distance class (m)	Probability of clonal identity	No. of pairs
1	1.000	7
2	1.000	32
3	0.833	36
4	0.429	21
5	0.083	12
6	0	11
7	0	18
8	0	27
9	0	25
10	0	28
11	0	21
12	0	29
13	0	31
14	0	25
15	0	22
16	0	11
17	0	11
18	0	5
19	0	5
20	0	1

around the stems to avoid disturbing the clumps. However, given the fact there was no dominant single stem for any of the three target clumps, in contrast to the root collar sprouting *Fagus engleriana* and *F. japonica*, which are commonly observed to have few large stems surrounded by many smaller stems in (Cao & Peters, 1998; Ohkubo, 1992), it is possible that most or all of the asexually derived stems grow via root sprouting or even from an underground lignotuber-like structure. However, if root sprouting is the main mode of clonal growth it must be over relatively short distances because of the close aggregation of clonal stems. It is possible that the formation of these large clumps via clonal growth was initiated via deformation of the original now lost trunk by wind and heavy snow (Homma, 1997). Although most of the stems are upright (Figure 3) and their growth is not associated with any obvious deformation of adjacent trunks, these could have senesced and decayed decades ago. Dead or nearly dead stems, most of which were lying at

acute angles near the ground surface, were observed in all three clumps, which is likely explained by exposure to persistent winter winds, typhoons (Nagashima et al., 2015) and/or heavy snow, weakening the connection of the stem to its base (Tanimoto, 1993). Layering may also have an important role in expanding the clumps given that some live stems were growing for a large portion of their length along the ground. Layering via adventitious roots has been observed in clonal individuals of *Fagus sylvatica* near the tree line in the Giant Mountains, Czech Republic (Vacek & Hejcman, 2012).

The underlying reason for clonal growth being important in the stand is likely related to the habitat being inhospitable for tree growth and survival of seedlings. The clumps grow on a relatively flat wide ridge which is at the limit of growth for forest trees in the area due to exposure to harsh winter winds (Mokudai & Koizumi, 2007). Despite relatively low elevation near the studied *F. crenata* stand (900–1,050 masl) the tree line is depressed in places due to persistent winter winds preventing tree growth and resulting in a patchwork of bare ground and alpine species-dominated grassland (Mokudai & Koizumi, 2007). The impact of wind on the growth form of *F. crenata* is well demonstrated by the fact that forest dominated by single-stemmed individuals occurs on the adjacent sheltered northwest facing slope (Figure 3d). Other factors that may favor clonal growth are the presence of *Sasa* bamboo, which can inhibit the establishment of *F. crenata* seedlings (Nakashizuka & Numata, 1982). However, this is likely to be a minor factor given that observations of potential examples of clonal growth are absent from areas where *Sasa* bamboo grows with *F. crenata* at less exposed sites. Another factor could be past cutting at the site, which in heavy snow regions has been observed to result in regeneration via a large number of adventitious shoots (Goto, Takahashi, & Ohtani, 2016). However, there was no evidence of recent cutting and the stem diameters vary considerably within a clump, suggesting continual regeneration of clonal stems rather than episodal – as would be expected as a result of cutting. Given this, it is unlikely that any past cutting could be the sole factor underlying the occurrence of clonal growth at the site.

5 | CONCLUSION

Overall, this study provides the first confirmation of clonal growth in *F. crenata* using genetic markers. Clonal growth is likely to be an important factor in the ability of the species to persist at high exposed elevations on the Osado Mountain Range of Sado Island and in other dwarf

beech forest on the snowy Japan Sea side of Honshu. Similar observations of clonal growth associated with heavy snow and exposure have been confirmed using genetic markers in other temperate trees on the Japan Sea side, including *Cryptomeria japonica* (Kimura, Nagashima, Kamitani, Sakio, & Tsumura, 2020) and *Quercus crispula* (Aizawa, Enkawa, & Ohkubo, 2017). It can be speculated that the capacity for clonal growth in *F. crenata* may explain how the species was able to maintain a wide but scattered distribution at low elevations in Honshu during the colder and drier Last Glacial Maximum when climatic conditions were likely to be highly marginal for the species (Tsukada, 1982). Further research is needed to uncover how the frequency of clonal growth differs with gradients in elevation, snow depth, exposure to winter winds and other environmental factors.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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